

ETHAN CHANG

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RESEARCH INTERESTS

Machine Learning, Astrocyte-Informed Neural Networks, Reinforcement Learning, Computational Neuroscience

EDUCATION

Washington University in St. Louis - *St. Louis, MO* 2025 - Present
Ph.D. in Neuroscience

Johns Hopkins University - *Baltimore, MD* 2025 - Present
M.S. in Computer Science

University of Rochester - *Rochester, NY* 2021 - 2025
B.A. in Mathematics and B.S. in Neuroscience, High Distinction. Minor in Psychology

RESEARCH EXPERIENCE

Washington University in St. Louis - *St. Louis, MO* 2025 - Present
Doctoral Researcher, Brain Dynamics and Control Research Group and Papouin Lab

- Co-advised by ShiNung Ching, Ph.D. (Department of Electrical and Systems Engineering) and Thomas Papouin, Ph.D. (Department of Neuroscience)
- Investigating computational models that incorporate astrocyte-inspired reinforcement learning architectures for understanding neural computation and adaptive behavior.

University of Rochester - *Rochester, NY* 2021 - 2025
Undergraduate Researcher, Nedergaard Lab (Maiken Nedergaard, M.D., D.M.Sc.)

- Conducted research on the role of aquaporin-4 localized on astrocytic endfeet in glymphatic transport dynamics.

PUBLICATIONS

- Gahn-Martinez D, Giannetto M, Chang E, Beam N, Pla V, Nedergaard M. Chronic intraventricular cannulation for the study of glymphatic transport. *eNeuro*. 2025. doi: 10.1523/ENEURO.0537-24.2025
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E, Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. *Glia*. 2024. doi: 10.1002/glia.24515

PRESENTATIONS

- Chang E (presenter), Giannetto M, Agarwald I, Vento N, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression on glymphatic clearance routes and function. Schwartz Discover Scholar Showcase. 2024.
- Chang E (presenter), Giannetto M, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression and size-dependent solute movement in the brain. University of Rochester Undergraduate Research Exposition. 2024.
- Chang E (presenter), Barth RK. Isolation of hydrogen sulfide-producing bacteria from the environment. Department of Microbiology and Immunology Poster Session. 2023.
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E (presenter), Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. University of Rochester Undergraduate Program in Biology and Medicine Poster Symposium. 2023.

AWARDS AND FUNDING

Washington University in St. Louis 2025 - 2027
NIH T32 Predoctoral Trainee, Neuroscience Training Program (T32NS121881)

University of Rochester 2024
Schwartz Discover Grant

TECHNICAL SKILLS

Selected Programming/ML: Python, Java, PyTorch, NumPy, pandas, scikit-learn, matplotlib

TEACHING

- Washington University in St. Louis, Department of Biology** 2026
Assistant in Instruction, BIOL 3411: Principles of the Nervous System
- Instructor: Keith Hengen, Ph.D and Mary Lambo, Ph.D.
- University of Rochester, Department of Brain and Cognitive Sciences** 2023
Teaching Assistant, NSCI 201P: Basic Neurobiology Lab
- Instructor: Renee Miller, Ph.D

SERVICE AND LEADERSHIP

- WashU Center for Career Engagement** 2025 - Present
Graduate Student Advisory Board Representative
- Provided graduate student input on career resources, especially for students interested in industry pathways.
- WashU St. Louis Neuroscience Outreach** 2025 - Present
Student Volunteer
- Participated in public neuroscience outreach at the Amazing Brain Carnival, St. Louis Science Center SciFest.
- University of Rochester Student Association** 2021 - 2025
Deputy Chair for Academic Affairs Committee
- Helped lead a team working on initiatives to improve the student academic experience and expand undergraduate research opportunities.
- University of Rochester Office of Undergraduate Research** 2023 - 2025
Student Research Ambassador
- Served as a student contact for research involvement and participated in admissions panels.
- Clinical and Emergency Service** 2021 - 2024
Strong Memorial Hospital; RC-MERT
- Emergency Department Clinical Support and Emergency Medical Technician roles in high-acuity care settings, including during the COVID-19 Omicron surge.

SELECTED COURSEWORK

AI/ML: Machine Learning; Advanced Machine Learning; Deep Learning; Probabilistic Graphical Models; Reasoning Under Uncertainty; Natural Language Processing

Math/Statistics: Probability; Mathematical Statistics; Stochastic Processes; Linear Algebra; Numerical Analysis; Multidimensional Calculus; Point-Set Topology

CS/Systems: Algorithms; Data Structures; Software Engineering; Computer Architecture; Computer Organization

LANGUAGES

English (native), Mandarin Chinese (conversational)